Afonso Paiva

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Portfolio: afonsopaiva.github.io/Portfolio

Summary

Backend engineer focused on Java and Spring Boot, with experience in REST APIs and microservices, and production experience in Go, Python and Angular. Author of Ariadne, an open-source JVM agent (v0.1.0-beta.1, JMH-validated, with OpenTelemetry and JMX observability) that restores the caller context lost across thread boundaries in asynchronous Java code. Founder of SkillBridge, a non-profit full-stack platform in production with 50+ weekly active users. Currently a Part-Time Backend Developer at INESC-TEC, working on the Python to Go backend migration.

Technical Skills

Languages: Java, Go, Python, SQL, JavaScript, C/C++, C#.

Backend: Spring Boot, Microservices, REST API design, JWT, Gin, Laravel, SQLAlchemy, GORM.

Frontend: Angular, JavaScript.

Data and Cloud: PostgreSQL, TimescaleDB, MySQL, Google Cloud Platform (GCP), Firebase, Stripe API.

DevOps and Practices: Docker, Git, Linux/Unix, CI/CD, Agile/Scrum, Unit Testing, JMH benchmarking, JaCoCo, SpotBugs.

Experience

Backend Developer (Part-Time) | INESC-TEC | August 2026 – Present

- Part of the team migrating INESC-TEC's backend from Python to Go.
- Leading the data layer redesign, preserving all foreign keys, constraints and relationships while configuring TimescaleDB hypertables.

Software Research Engineering Intern | INESC-TEC (Robotair UAV platform) | June 2026 – August 2026

- Architected modular ROS/Docker components for the Robotair UAV platform, enabling air-gapped deployment of 5+ containerized services for secure field operations.
- Wrote technical documentation and contributed to CI/CD pipeline configuration for reproducible, air-gapped deployment workflows.

Software Engineering Intern | YouOn | February 2026 – May 2026

- Engineered 3 internal dashboards serving 15+ team members, reducing manual data retrieval time by about 40%.
- Built reusable Angular components consumed by 2 client-facing products.
- Designed MySQL schemas and Laravel API routes for relational data management, supporting real-time data pipelines across 3 internal tools.

Projects

Ariadne: Async Causality Tracing for the JVM (Java 21, ByteBuddy, Reactor, JMH) | GitHub | Website

- Built an open-source, zero-code Java agent (Apache 2.0) that restores the caller stack trace lost at thread boundaries (ExecutorService, ForkJoinPool, CompletableFuture, virtual threads, Reactor, RxJava) and attaches it to failures as an AsyncCausalityException via addSuppressed, with optional SLF4J MDC propagation.
- Published v0.1.0-beta.1 with W3C TraceContext propagation, an OpenTelemetry bridge, a Kotlin Coroutines adapter, a JDK 21 Scoped Values carrier and JMX observability.
- Designed a lazy, low-overhead model: a 40-byte immutable link per dispatch, allocation in under 4 ns, and stack synthesis only when an exception occurs. Made it production-safe with fail-safe advice, a runtime kill switch, memory bounded to O(maxDepth) that always preserves the root origin, and an MDC key allowlist.
- Validated with JMH on OpenJDK 21: about 1.5% overhead on Reactor Mono hops with 5.8x less allocation than Hooks.onOperatorDebug, and 14.2x faster with 9.2x less allocation per trace node than an OpenTelemetry span. Backed by 100+ tests, including a forked-JVM javaagent integration test, with CI running JaCoCo and SpotBugs.

SkillBridge (Go, Gin, PostgreSQL, Angular, JWT) | Website

- Founded and built this free non-profit academic platform, featured publicly on RTP, serving 50+ weekly active users.
- Built and deployed the production backend on Google Cloud Platform, including matchmaking logic, JWT-secured APIs and peer-review data systems.

Solar Terrain Analytics (Java 21, Spring Boot, Google Maps API, PostgreSQL) | GitHub

- Developed a Spring Boot REST API orchestrating geospatial data from the Google Maps API, persisted in PostgreSQL.
- Optimized the ROI calculation engine, improving query processing speed by 65% while handling 50000+ data points, and designed polygon-based analysis algorithms with ROI projections based on energy costs.

WebCommerce-BookStore (Java 21, Spring Boot, Microservices, Stripe, REST API) | Website

- Architected and deployed on Google Cloud Platform a full-stack e-commerce platform with a Spring Boot microservices backend, a secure private REST API managing relational data for writers, books and clients, and Stripe integration for payments.

LoadX (C++, OpenGL, GLSL) | GitHub

- Custom graphics engine with real-time Physically Based Rendering (PBR), running at a consistent 60 FPS.

Education

M.Sc. in Computer Engineering | ISEP, Porto | Expected 2028

B.Sc. in Digital Games Engineering | IPCA, Braga | Graduated June 2026

Coursework: Object-Oriented Programming, Data Structures, Databases, Computer Vision, Linear Algebra.

Erasmus+ Exchange | HOWEST University of Applied Sciences | September 2025 – January 2026: software design patterns and object-oriented architecture in C#.

Blended Intensive Programme | Nord universitet | March – April 2025: mixed reality research project in Unity, in an international, English-taught cohort.

Piscine Program | 42 Porto | 2023

Spoken languages: Portuguese (native), English (B2, Cambridge certificate), Spanish (B1), Italian (A1)